MATH 243: Discussion Worksheet 11

Topics: 15.7 Triple Integrals in Cylindrical Coordinates; 15.8 Triple Integrals in Spherical Coordinates

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1. Rewrite each equation in cylindrical coordinates and identify the surface.
   (a) $x^2 + y^2 = 4$
   (b) $z = 4 - x^2 - y^2$
   (c) $x^2 + y^2 + z^2 = 9$

2. Find the volume of each solid by computing $\iiint_E dV$ using cylindrical coordinates.
   (a) The solid in the first octant that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, and the planes $z = 0$ and $z = 4$.
   (b) The solid that lies inside the cylinder $x^2 + y^2 = 4$, above the plane $z = 0$, and below the plane $z = 3 + x$.
   (c) The solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$.

3. Set up a triple integral using cylindrical coordinates that gives the volume of the solid inside the sphere $x^2 + y^2 + z^2 = 2$ and above the paraboloid $z = x^2 + y^2$. *Do not evaluate.*

4. Rewrite the following iterated integral using cylindrical coordinates. *Do not evaluate.*

$$\int_{-3}^{3} \int_{0}^{\sqrt{9-x^2}} \int_{0}^{9-x^2-y^2} \sqrt{x^2+y^2}\, dz\, dy\, dx$$

5. Using spherical coordinates, evaluate the triple integral

$$\iiint_E (x^2 + y^2 + z^2)\, dV,$$

   where E is the solid region between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.

6. Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the xy-plane, and below the cone $z = \sqrt{x^2 + y^2}$. Use spherical coordinates.

7. Use spherical coordinates to find the volume of the solid in the first octant that lies inside the sphere $x^2 + y^2 + z^2 = 9$ and above the plane $z = 1$.

8. Evaluate the following integral by switching to spherical coordinates.

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}} (x^2 + y^2 + z^2)^{3/2}\, dz\, dy\, dx$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Cylindrical Coordinates. In the cylindrical coordinate system, a point P in space is represented by an ordered triple (r, θ, z), where (r, θ) gives the polar coordinates of the projection of P onto the xy-plane, and z is the same z-coordinate as in rectangular coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

If a solid E can be described in cylindrical coordinates by

$$E = \{(r, \theta, z) \mid \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta),\ u_1(r, \theta) \le z \le u_2(r, \theta)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r,\theta)}^{u_2(r,\theta)} f(r\cos\theta, r\sin\theta, z)\, r\, dz\, dr\, d\theta.$$

Cylindrical coordinates can also be defined using polar coordinates in the yz-plane or the xz-plane instead of the xy-plane, and in each case the Jacobian factor is still r.

Spherical Coordinates. In the spherical coordinate system, a point P in space is represented by an ordered triple (ρ, θ, ϕ), where ρ is the distance from the origin to P, θ is the same angle as in cylindrical coordinates, and ϕ is the angle between the positive z-axis and the line segment from the origin to P:

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

If a solid E can be described in spherical coordinates by

$$E = \{(\rho, \theta, \phi) \mid a \le \phi \le b,\ h_1(\phi) \le \theta \le h_2(\phi),\ u_1(\theta, \phi) \le \rho \le u_2(\theta, \phi)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_a^b \int_{h_1(\phi)}^{h_2(\phi)} \int_{u_1(\theta,\phi)}^{u_2(\theta,\phi)} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi\, d\rho\, d\theta\, d\phi.$$

Suggested Textbook Problems
Section 15.7: 1-32
Section 15.8: 1-32,37-40,43-46